

Introduction

DSToolkit is an interactive desktop application developed in Java designed to bridge the gap between abstract data structure concepts and practical understanding. By providing real-time visual representations of complex algorithms, the toolkit helps students and developers visualize how data is stored, moved, and manipulated within memory.

Project Objectives

- To provide a graphical user interface (GUI) for various data structures.
- To demonstrate the working mechanisms of Binary Search Trees (BST), Linked Lists, and Queues.
- To visualize pointers, node addresses, and traversal paths (Inorder, Preorder, Postorder).
- To simplify the learning curve for beginners studying Data Structures and Algorithms (DSA).

Technology Stack

Language: Java

GUI Framework: Java Swing and AWT (Abstract Window Toolkit)

Graphics: Java 2D API (Graphics2D) for custom node rendering and animations

Concepts Used: Object-Oriented Programming (OOP), Recursion.

Core Modules & Features

- A. Binary Search Tree (BST) Visualization (Class: BSTVisualization.java)**
 - Dynamic insertion visualization using recursion.
 - Real-time traversal display (Inorder, Preorder, Postorder).
 - Timer-based animations for node highlighting.
- B. Doubly Linked List (DLL) Visualization (Class: DoublyLinkedListVisualization.java)**
 - Bidirectional pointer visualization.
 - Simulated memory address display (e.g., 0x7A2B).
 - Head and tail insertion visualization.
- C. Priority Queue Visualization (Class: PriorityQueueVisualization.java)**
 - Priority-based sorting (small to large).

- Front and Rear pointer indicators.
- Capacity configuration with overflow visualization.

Implementation Details

The project utilizes custom Node structures within each visualization class.

- BST Node: Contains left and right pointers.
- Doubly Linked List DNode: Contains next, prev pointers and a simulated address field.
- Graphics2D: Used to render RoundRectangle2D shapes for nodes with a clean modern UI.

Conclusion

DSToolkit successfully implements a functional and educational environment for visualizing data structures. The Java Swing-based GUI provides a responsive desktop experience, while custom graphics logic makes abstract pointer and tree operations visually understandable.